

Design and Development of a Human-Electric Hybrid Canal Cleaning System

Final Project Report

Submitted by

Team 9

Tadouanla Guetchuin Billy	28177
Taosif Wasi Eshan	38607
Tifang Ngan Tata	38181
Muhammad Abubakar	38247

Engineering for Sustainability (B.Sc)

Hochschule Rhein-Waal
Faculty of Technology and Bionics

July, 30th 2026

Contents

Introduction	2
Background & Statement of Problem	2
Background	2
Statement of Problem	3
Requirements list and Stakeholders	3
1. Functional Design	4
Core Principles	4
2. Utility Value Analysis (<i>Nutzwertanalyse</i>)	4
Methodology	4
The 5-Step Evaluation Process	4
Evaluation Scale Systems	5
Methodological Limitations	5
3. Key Takeaways	5

Introduction

Background & Statement of Problem

Background

Water is the most valuable natural resource and plays a vital role in mainly maintaining a healthy ecosystem, supporting biodiversity. Freshwater bodies such as **rivers**, **lakes**, and **canals** are home to numerous aquatic organisms. However, these water bodies are increasingly threatened by pollution, climate change, and excessive nutrient inputs, which can lead to environmental problems such as harmful algal growth.

Algae are naturally occurring photosynthetic organisms found in freshwater and marine ecosystems. Under normal conditions, they are an essential part of aquatic ecosystems because they produce oxygen through photosynthesis and form the base of many aquatic food chains. However, when Nutrients such as **Nitrogen** and **Phosphorus** become excessively available due to agricultural **runoff**, **wastewater discharge**, **urban stormwater** or other human activities, algae can multiply rapidly. Combined with warm temperatures, sufficient sunlight, and slow-moving water, these conditions often result in excessive algal growth, commonly known as **algal blooms**.

Excessive algae growth has many negative **environmental**, **social** and **economic consequences**. Thick layers of algae reduce the penetration of sunlight into the water, preventing submerged aquatic plants from carrying out photosynthesis. As algae die and decompose, microorganisms consume large amounts of dissolved oxygen, leading to oxygen depletion that can stress or kill fish and other aquatic organisms. In severe cases, algal blooms may produce **unpleasant odours**, **reduce water clarity**, **clog waterways**, **interfere with recreational activities**, and significantly decrease the aesthetic value of natural environments.

The **Spoey Canal in Kleve, Germany**, Germany, experiences recurring algae growth, making it an important environmental challenge for the city. Besides being part of the local drainage and water management system, the canal is an attractive feature of residents and visitors and contributes to the city's landscape. Because the canal passes through public areas, maintaining its **ecological health and visual appearance is essential**.

As part of this initiative, **Hochschule Rhein-Waal (HSRW)** collaborates with the city of Kleve by involving engineering students in solving real-world sustainability challenges. Instead of simply proposing theoretical ideas, students apply the **VDI2221 Engineering design methodology** as well as the **VDI2220 Engineering design methodology**, which emphasizes systematic product development. The design process begins by understanding the **problem**, **analysing stakeholders**, **identifying customer needs**, and translating these needs into measurable engineering requirements before generating design

concepts.

In this project, students are expected to develop a **sustainable engineering solution** capable of reducing algae accumulation in the **Spoy Canal** while balancing technical *performance, environmental protection, economic feasibility, safety, maintainability, and sustainability*.

Statement of Problem

The Spoy Canal currently experiences recurring algae accumulation, which negatively affects the ecological conditions and visual quality of the waterway. Excessive algae reduce water quality, decrease dissolved oxygen concentrations, interfere with aquatic ecosystems, and create an unpleasant environment for both residents and visitors. The Problem also increases maintenance efforts and cost of the local authorities responsible for managing the canal.

As Kleve prepares for **LaGa 2029**, the existing algae problem presents additional challenges. A canal covered with algae is inconsistent with the objectives of **LaGa**, which seeks to promote **environmental sustainability, biodiversity and attractive urban landscape**. Therefore, an effective and sustainable solution is required to improve the condition of the canal before and during the event.

Traditional algae removal methods often require considerable **manual labour, repeated maintenance, and operational costs**. Some methods may also disturb aquatic habitats or consume significant amounts of energy. Consequently, there is a need for an innovative engineering solution that removes algae efficiently while minimizing environmental impacts.

The proposed solution should not only remove algae effectively but also operate safely in public areas. Consume little energy, protect aquatic organisms, resist corrosion in freshwater environments, require minimal maintenance, and comply with environmental and safety regulations. Since the machine will potentially operate in areas visited by the public during **LaGa 2029**, it should also have an **aesthetically acceptable appearance and produce minimal noise**.

Following the engineering design methodology, the first stage of the project is not to select a specific but to define what the product must achieve. This involves identifying stakeholder needs and translating them into measurable engineering requirements that will guide concept generation and evaluation in the following stages of the project.

Requirements list and Stakeholders

The overall engineering challenge is therefore to design a sustainable, reliable, safe, and efficient algae-removal machine capable of improving the ecological condition of visual quality of the **Spoy Canal** while satisfying technical, environmental, economic, and stakeholder requirements. Such a solution would contribute to the

long-term management of the canal and support the environmental objectives of **LaGa 2029**.

1. Functional Design

Core Principles

Functional Design defines **what** a system must do before establishing physical solutions or technical mechanisms.

- **Solution Neutrality:** Functions are defined without committing prematurely to specific technologies or physical components.
- **System Boundaries & Flows:** The overall function maps system inputs into outputs across three primary categories:
 - **Energy (E):** Electrical, mechanical, or thermal energy transfers.
 - **Signal (S):** Control inputs, sensor data, or status feedback.
 - **Mass / Matter (M):** Materials, fluids, or workpieces passing through the system.
- **Function Categories:** Core operational actions used to structure sub-functions:
 - *Conversion* (changing input form to output form)
 - *Varying* (amplifying, stepping down, or modulating)
 - *Connecting/Disconnecting* (switching, coupling, or mixing)
 - *Channeling* (directing or transferring power, signals, or mass)
 - *Storing* (holding energy, signals, or mass)

2. Utility Value Analysis (*Nutzwertanalyse*)

Methodology

Utility Value Analysis is a quantitative multi-criteria decision-making framework used to evaluate, rank, and select competing design variants.

The 5-Step Evaluation Process

1. **Define Alternatives:** Identify concept variants generated during conceptual design.
2. **Select Evaluation Criteria:** Derive criteria directly from the system requirements list (e.g., performance, safety, manufacturability).
3. **Weight Criteria:** Assign relative weight factors (w_i) to criteria such that $\sum w_i = 1.0$ (or 100%).
4. **Score Alternatives:** Rate performance (v_i) of each concept against each criterion using a standardized rating scale.
5. **Calculate Overall Utility Value:** Compute total weighted utility:

$$\text{Utility Score} = \sum_{i=1}^n (v_i \cdot w_i)$$

Evaluation Scale Systems

- **Standard Use-Value Analysis (0 to 10 Scale):** Evaluates performance on an 11-point continuum where 0 represents *Useless*, 5 represents *Satisfactory*, and 10 represents *Ideal*.
- **VDI 2225 Guideline (0 to 4 Scale):** Standardizes rating across 5 discrete levels: 0 (*Unsatisfactory*), 1 (*Just tolerable*), 2 (*Adequate*), 3 (*Good*), and 4 (*Ideal / Very good*).

Methodological Limitations

- **Subjectivity:** Rating scores can be influenced by team bias or subjective interpretations.
 - **False Precision:** Mathematical scoring creates an illusion of objective accuracy.
 - **Sensitivity:** Final rankings can shift significantly with minor changes to weighting distributions.
-

3. Key Takeaways

- **Solution-neutral modeling** prevents premature design bias and clarifies fundamental system transformations.
- **Multi-criteria evaluation** transforms qualitative, multi-variable design trade-offs into structured, documentable decisions.